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Enhanced high-voltage cycling stability of Ni-rich $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ cathode coated with Li_2O-$2B_2O_3$

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Abstract

$LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM811) is considered one of the most competitive cathode materials for lithium-ion batteries for use in next-generation electric vehicles. However, the poor cycle stability of layered Ni-rich cathode materials at high voltages (>4.3 V) has hindered their commercial development. In this study, we evaluate the effects of lithium boron oxide (LBO) glass as a lithium-ion-conductive coating layer on the structure, morphology, and electrochemical properties of NCM811 at a high cutoff voltage of 4.5 V. We show that NCM811 coated with 0.3 wt% LBO exhibits improved cycling retention (82.1%) compared with bare NCM811 (50.8%) after 100 cycles at 1C. Electrochemical impedance spectra and scanning electron microscopy and transmission electron microscopy images after cycling reveal that the LBO-coating layer not only lowered the charge-transfer resistance between the electrode and electrolyte but also improved the structural stability during cycling.

Keywords:

Ni-rich cathode, lithium boron oxide glass, high voltage, surface coating, cycling stability

1. Introduction

Today, the automotive industry faces increasingly stringent requirements in terms of fuel economy and emissions reduction. Significant attention has thus been focused on rechargeable lithium-ion batteries for use in electric and electric hybrid vehicles [1, 2]. Next-generation electric vehicles with a driving range of at least 300 miles will penetrate the mass consumer market in the immediate future [3]. As the cathode remains the limiting component in state-of-the-art lithium-ion batteries, substantial efforts have been focused on the development of low-cost, high-energy-density, and high-rate-capability cathode materials [4-6]. Recently, layered Ni-rich lithium transition-metal oxides have appeared as promising candidates; increasing the nickel content leads to a higher specific capacity of up to approximately 200 mAh g^{-1} for NCM811 [7, 8]. To meet practical demands, researchers have suggested further increasing the lithium extraction from Ni-rich cathodes under redox potentials of up to 4.5 V [9, 10]. However, the resulting poor cycling stability and rate capability at high voltage may hinder the success of this approach [11, 12]. Many attempts have been made to understand the failure mechanism of Ni-rich cathodes; challenges have been shown to arise from (i) undesirable phase transformations from the layered structure to a spinel and even a rock-salt phase with prolonged Li deficiency near the surface region of Ni-rich cathodes at high voltages [13-15]; (ii) dissolution of metal ions upon directly exposing NCM cathode materials to $LiPF_6$-based organic electrolytes [16, 17]; and (iii) the thicker surface film resulting from electrolyte decomposition impeding the diffusion of lithium ions and leading to an increase of the internal resistance [18]. These factors indicate that the interface between the electrode and electrolyte is the crucial area for Li^+ diffusion, surface reconstruction, and side reactions.

Researchers have made extensive efforts to avoid these adverse issues and make full use of the virtues of Ni-rich cathodes. Among the available strategies, surface coating is believed to be a facile and effective way to improve the electrochemical performance of NCM cathode materials, and conventional approaches can generally be divided into two categories. The first category of approaches involves the use of metal oxide and fluoride coating layers such as ZrO_2 [19], Al_2O_3 [20], Y_2O_3 [21, 22], and AlF_3 [23], which serve as excellent protective layers but decrease the lithium-ion transfer rate at the interface because of their inactive nature electrochemically and in terms of electrical conductivity. The other category of approaches involves the use of lithium-ion-conductive coating layers, such as Li_2TiO_3 [24], $Li_4Ti_5O_{12}$ [25], Li_3PO_4 [26], Li_3VO_4 [27], and Li_2ZrO_3 [28, 29], which is a step forward to address the problem; however, control of the thickness and uniformity of the coating layer remains far from satisfactory.

It has recently been shown that Li_2O-$2B_2O_3$ (LBO) glass, a fast ionic conductor with good lithium-ion conductivity, easily forms a homogeneous coating on the surface of Ni-rich cathodes because of its good wetting property and relatively low viscosity [30-32] and appears to be a good solution. In addition, electrochemical studies have shown that LBO glass materials are stable at high oxidation potentials in high-voltage lithium-ion batteries [33, 34]. Ying et al. [35] reported that $LiNi_{0.8}Co_{0.2}O_2$ coated with LBO exhibited remarkably enhanced electrochemical reversibility at elevated temperature (60 °C). Chen et al. [36] coated $Li_{1.2}Ni_{0.2}Mn_{0.6}O_2$ with LBO, stabilizing the structure and lowering the interface resistance at low temperature (-30 °C). On the basis of these findings, we assumed that the presence of LBO glass on the surface of host materials may contribute to improving the electrochemical performance even at high voltages. Nonetheless, the behavior of LBO-glass-coated $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ at high voltage has not yet been investigated.

In this work, we proposed and designed a facile technique to coat LBO glass on NCM811 using a simple wet-chemical process. The effects of the coating layer on the structure, morphology, and electrochemical performance at a high cutoff voltage of 4.5 V are discussed in detail. The electrochemical impedance spectra and scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images after cycling confirmed that a LBO coating of adequate thickness serves as an effective protective layer to improve the electrochemical performance.

2. Experimental section

2.1 Material synthesis

The $Ni_{0.8}Co_{0.1}Mn_{0.1}(OH)_2$ precursors were synthesized via the oxalate coprecipitation method by using a 2 L continuous stirred tank reactor (CSTR). $NiSO_4 \cdot 6H_2O$, $CoSO_4 \cdot 7H_2O$ and $MnSO_4 \cdot H_2O$ (molar ratio of Ni: Co: Mn = 8:1:1, analytical grade) were dissolved in 200 mL deionized water to obtain a 2 M solution. 100 mL of 2 M $NH_3 \cdot H_2O$ was added into the reactor as base solution and heated to 50 °C. Subsequently, the metal solutions were added dropwise into the reactor under a N_2 atmosphere. At the same time, a 2.0 M NaOH solution and the desired amount of $NH_3 \cdot H_2O$ solution as a chelating agent was separately pumped into the reactor. The co-precipitation reaction temperature and pH were maintained at 50 °C and 11, respectively, with a stirring speed of 600 rpm. After the 12 h reaction time, the precursor was then filtered, washed, and dried in a vacuum oven at 120 °C for 12 h. The $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ was synthesized by mixing the precursors with a stoichiometric amount of $LiOH \cdot H_2O$ (99% Ar) with a molar ratio of 1:1.05 (5% excess of $LiOH \cdot H_2O$); then, the mixture powder was annealed at 500 °C for 5 h and 800 °C for 12 h at 5 °C min^{-1} under flowing oxygen.

LBO-glass-coated $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ was prepared using a solution method. The precursor of LBO glass was synthesized by dissolving $LiOH \cdot H_2O$ and H_3BO_3 (1:2 molar ratio) in ethanol under strong stirring until the components were entirely dissolved. Then, the $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ powders were added into the solution and gently heated to 60 °C. When the solvent completely evaporated, the powders were calcined at 500 °C for 10 h at 5 °C min^{-1}. The mass contents of the LBO-coated NCM811 samples were set to 0.2, 0.3, and 0.4 wt%, and the samples are denoted hereafter as LBO 0.2, LBO 0.3, LBO 0.4, respectively. The exact amounts of precursors for LBO coating samples were shown in Table S1.

2.2 Material characterization

The crystalline phases of each sample were determined using a Philips X-ray diffractometer (APD 3520) using Cu Kα radiation in the 2θ range of 10°-80°. The compositions of the as-synthesized samples were measured using inductively coupled plasma atomic emission spectroscopy (ICP-AES; Optima 5300 DV). The morphologies and surface compositions of the pristine NCM811 and LBO-coated samples were evaluated using SEM (S-4800) at an operating voltage of 5 kV, energy-dispersive X-ray spectrometry (EDS; Strata FIB 201), and TEM (TECNAI G^2 F20).

2.3 Electrochemical measurements

All the half-cell electrochemical measurements were performed using CR2025-type coin cells. Composite cathode electrodes were prepared by thoroughly mixing the cathode powder, Super-P, and poly (vinylidene fluoride) binder (weight ratio of 8:1:1) in N-methyl-2-pyrrolidone solution and pasting on an aluminum-foil current collector followed by drying at 120 °C for 12 h in a vacuum oven. All of the half cells were assembled in an Ar-filled glove box with 1 M $LiPF_6$ in an ethylene carbonate (EC)-diethyl carbonate (DEC) mixture (1:1 volume ratio) as the electrolyte, and a polypropylene microporous film was used as the separator. The galvanostatic charge-discharge characteristics (1C = 180 mA g^{-1}) were evaluated on a BTS series battery testing system (Neware Technology) in the voltage range of 2.75-4.5 V at room temperature. Cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) were performed on a CHI760E electrochemical workstation. For the CV tests, cycling was performed between 2.75 and 4.5 V (vs. Li/Li^+) at 0.1 mV s^{-1}. The EIS measurements were conducted over a frequency range from 0.01 Hz to 100 kHz.

Results and discussion

3.1 Physical characteristics

XRD patterns of the pristine NCM811, LBO 0.2, LBO 0.3, and LBO 0.4 samples are presented in Fig. 1. All the diffraction peaks for NCM811 correspond well to a layered α-$NaFeO_2$ structure (space group $R\text{-}\bar{3}m$) without obvious impurities. The marked splitting of the (006)/(012) and (018)/(110) diffraction peaks of all the samples indicate that a well-developed layered structure was realized [37, 38]. There are no obvious differences in the XRD patterns of the pristine NCM811 and

LBO-coated samples, suggesting that the coating process had almost no effect on the NCM 811 crystal structure. This result may stem from the low concentration or amorphous structure of LBO glass not being detectable by XRD.

To confirm whether the coating layer formed on the surface of the pristine NCM811, SEM was used to analyze the morphology of all the samples; the results are presented in Fig. 2. All the samples exhibited a quasi-spherical morphology (Fig. 2a, c, e, g), formed by the accumulation of many polyhedral particles with diameters of approximately 200-300 nm (Fig. 2b, d, f, h). It is worth noting that only the pristine NCM811 exhibited clear and smooth surfaces (Fig. 2b). The surface morphology became increasingly rougher with increasing LBO coating amount (Fig. 2d, f, h). Furthermore, the elemental distribution of the LBO glass or NCM811 surface was characterized using EDS mapping, as shown in Fig. S1 (NCM811) and Fig. 3 (LBO 0.3). In the selected region of Fig. 3, the elemental mappings of Ni, Co, Mn, O, and B are homogeneously distributed, indicating that the LBO glass was evenly coated on the pristine NCM811 microspheres. In addition, the ICP results presented in Table S1 suggest that the measured molar ratio of Li, Ni, Co, Mn, and B was very close to the theoretical value.

We also evaluated the surfaces of the pristine NCM811 and LBO 0.3 samples using FE-TEM; the results are presented in Fig. 4. The pristine NCM811 had a clear and smooth surface (Fig. 4a), whereas LBO 0.3 had a blurry and rough edge (Fig. 4c). Closer inspection of the pristine NCM811 sample in Fig. 4b revealed an interplanar distance of 0.476 nm, which corresponds to the (003) planes of $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$. After coating with LBO glass, a thin-film surface coating layer was observed on the LBO 0.2 (Fig. S2a), LBO 0.3 (Fig. 4d), and LBO 0.4 (Fig. S2b) samples unlike for the pristine NCM811 (Fig. 4b). This result indicates that the coating structure was formed on the surface of the NCM811. The thickness of the coating layers increased with increasing amount of LBO glass. Therefore, the coating layer thickness should be optimized to better prevent the cathode material from being attacked by the electrolyte. Of the tested samples, the LBO 0.3 sample with a thickness of approximately 7 nm encapsulated the host material perfectly.

3.2 Electrochemical performance

To explore the effect of the LBO coating layer on the high-voltage electrochemical properties, galvanostatic charge/discharge tests at 0.1C (18 mA g^{-1}) were performed at 4.5 V. The initial discharge capacities of the pristine NCM811, LBO 0.2, LBO 0.3, and LBO 0.4 samples were 202.5 mAh g^{-1} (85.70%), 199.2 mAh g^{-1} (86.47%), 207.8 mAh g^{-1} (87.80%), and 204.9 mAh g^{-1} (87.21%), respectively (Fig. 5a). The pristine and LBO-coated NCM811 samples exhibited similar electrochemical behavior. All the LBO-coated samples appeared to have a higher coulombic efficiency than the bare NCM811, and the discharge capacities of the LBO 0.3 and LBO 0.4 samples were slightly higher than that of the pristine sample. These results may have been observed because an appropriate LBO coating can promote the delivery of lithium ions on the surface, which in turn increases the reversible capacity, as previously reported [31, 35, 39, 40].

However, NCM811 exhibits poor cycling stability when charging to high potentials because of the oxidation of the electrolyte and side reactions between the cathode and electrolyte [41]. To further demonstrate the possible effect of the LBO glass coating at higher voltage, the cycling stability of the pristine NCM811 and LBO-coated samples was examined between 2.75 and 4.5 V at a rate of 1C, as shown in Fig. 5c. The pristine NCM811 sample exhibits severe capacity fade from 189.1 to 96.0 mAh g^{-1} with a capacity retention of only 50.8% after 100 cycles. The LBO 0.2, LBO 0.3, and LBO 0.4 have capacity retention values of 71.8% (from 188.5 to 135.3 mAh g^{-1}), 82.1% (from 192.0 to 157.7 mAh g^{-1}), and 75.9% (from 189.0 to 143.5 mAh g^{-1}), respectively (Fig. 5c, d). These results suggest that the surface of the pristine NCM811 was severely damaged when charged to 4.5 V and that the LBO coating significantly improved the high-voltage cycling stability of NCM 811. As we know, the cathode material is a quasi-sphere formed by the accumulation of many polyhedral particles, and the connection between these particles is more easily eroded by the electrolyte, especially at high voltage [42]. However, the protective film on the surface after the LBO glass coating prevents direct exposure of the cathode materials to the acidic liquid electrolyte.

LBO is known to be a good lithium-ion conductor and is often used as the solid electrolyte in all-solid-state lithium-ion batteries [32]. The LBO glass interfacial coatings had a positive effect on the transfer of Li^+ on the surface of NCM811 electrodes and resulted in good rate capability compared with pristine NCM811, as shown in Fig. 5b. First, the cells were charged and discharged at 0.1C for three cycles to stimulate the batteries, followed by cycling at 0.2C, 0.5C, 1C, 2C, 5C, and 0.2C for every five cycles, respectively, in the voltage range of 2.75-4.5 V. The LBO 0.2 and LBO 0.3 samples exhibited better rate capabilities than the pristine NCM811, especially at high C-rates. This finding indicates that the moderate LBO glass coating layer effectively facilitates Li^+ migration in the cells.

To further understand the electrochemical behavior of the electrode materials, the first three cycles of the cyclic voltammograms for the pristine NCM811 and LBO

0.3 samples were recorded at a scan rate of 0.1 mV s^{-1} in the voltage range of 2.75-4.5 V, as shown in Fig. 5e, f. The redox peaks of the two samples have similar shapes, indicating that the LBO coating layers do not affect the electrochemical reaction of NCM 811. There are two distinct redox peaks with a sharp oxidation peak at 3.944 V and a small oxidation peak at 4.225 V during charging because of multiphase transitions for NCM811 [24]. The corresponding reduction peaks appear at 3.720 and 4.175 V. It is worth mentioning that the potential gap of LBO 0.3 (0.192 V) between the first anodic and cathodic peak is smaller than that of NCM811 (0.224 V), implying that the LBO coating is beneficial for reducing the electrochemical polarization and improving the electrochemical performance.

EIS spectra were obtained to further investigate the possible reasons for the enhanced electrochemical performance of the LBO-coated NCM811 samples after the 1st and 100th cycles in the voltage range of 2.75-4.5 V then charged to 3.8 V (1C) in the frequency range from 0.01 Hz to 100 kHz, as shown in Fig. 6a and b; the insets show the fitted equivalent circuits. All the impedance spectra consisted of a semicircle at high frequency, a semicircle at medium frequency, and an oblique line at low frequency. In the fitted equivalent circuit, the intercept at high frequency corresponds to the electrolyte resistance (R_s), the first semicircle in the high-frequency region corresponds to the resistance R_f (solid electrolyte interface (SEI) film resistance), and the second semicircle in the medium-frequency region corresponds to the charge-transfer resistance at the electrode/electrolyte interface, R_{ct}. CPE is the capacitance of the electrode/electrolyte double layer, and the Warburg impedance (Wo), corresponding to the oblique line in the low-frequency region, is related to solid-state diffusion of Li^+ in the bulk material. The values of R_f and R_{ct} for all the samples are listed in Table S2. R_{ct} of the pristine NCM811 was larger than the values of the LBO-coated samples, especially after 100 cycles, and R_f of LBO 0.3 was the lowest of all the samples, suggesting an enhancement of the kinetics of Li^+ diffusion through the interface and a charge-transfer reaction as well as a smaller electrode polarization after appropriate LBO coating. However, the thickness of LBO 0.2 seems too thin to encapsulate the NCM811 cathode material very well during the cycling (shown in fig. 2), and the LBO coating for LBO 0.4 seems so thick that it limited the rate of electronic delivery and Li^+ transport [36]. In addition, the sharp increase in R_{ct} of the pristine NCM811 after 100 cycles correlates with the distortion of the surface induced by side reactions with the electrolyte, which is consistent with the change in the surface morphology compared with the LBO 0.3 (Fig.S3). As can be seen from the further enlarged images in Fig. 7, the quasi-spherical morphology of the pristine NCM811 (Fig. 7a and b), which accumulated as many polyhedral particles became isolated with large cracks among the particles; in contrast, the morphology of LBO 0.3 (Fig. 7c and d) after 100 cycles appeared similar to the original morphology. These findings indicate that the LBO-coated layer is effective for resisting electrolyte decomposition and protecting the surface from attack by the electrolyte, leading to improved surface stability of the NCM811.

It is well known that the surface lattice structures of Ni-rich cathodes undergo an irreversible transformation during cycling under high-voltage operation [43]. To explore the effects of the LBO glass coating on the structure of NCM811, XRD spectra of the pristine NCM811 and LBO 0.3 electrodes after 100 cycles were obtained (Fig. 8). As observed in Fig. 8a, the intensities of all diffraction peaks of both samples became weaker after 100 cycles, indicating that the layered crystal structure of both electrodes suffered varying degrees of damage. In addition, the (003) peak of the LBO 0.3 sample showed almost no shift compared with that of the pristine sample after cycling, as shown in Fig. 8b. Moreover, the (006)/(102) and (108)/(110) peaks of the pristine NCM811 vanished after 100 cycles (Fig. 8c, d). These findings further confirmed that an appropriate amount of LBO coating is beneficial for stabilizing the structure of NCM811 cathode materials at high voltage. In addition, high-resolution TEM (HR-TEM) and fast Fourier transform (FFT) images were obtained to further determine whether local structure changes occurred; the results are presented in Fig. 8. Diffraction spots from the spinel phase were detected in the region marked by the red box for the pristine NCM 811 sample, as shown in Fig. 8e. For the LBO-coated samples, interfacial reaction with the electrode was prevented, and a layered structure was maintained (Fig. 8f). This finding indicates that the LBO coating layer not only retards interfacial reactions but also enhances the electrode surface stability to improve the cycling performance.

4. Conclusion

In summary, the layered Ni-rich cathode material NCM811 was successfully synthesized using an oxalate coprecipitation method followed by coating with different amounts of LBO glass (0.2, 0.3, and 0.4 wt%) via a simple heat treatment. Electrochemical characterization revealed that the 0.3 wt% LBO-coated sample exhibited a higher cycling retention (82.1%) than the bare NCM811 (50.8%) after 100 cycles. XRD, SEM, and TEM analyses showed that the LBO coating layer was well-developed on the surface of the pristine NCM811, preventing the cathode material from being directly exposed to the acidic liquid electrolyte and retarding the dissolution of transition-metal ions and HF attack, thereby improving the cycling stability at the high voltage of 4.5 V. In addition, the EIS results suggest that an appropriate amount of LBO coating, as a good Li^+ conductor, can reduce the charge-transfer resistance of the electrodes and enhance the rate capability compared with a metal oxide coating. This work reveals that the utilization of lithium-ion conductive glass with good wetting properties provides an opportunity to construct high-performance cathode materials for lithium-ion batteries.

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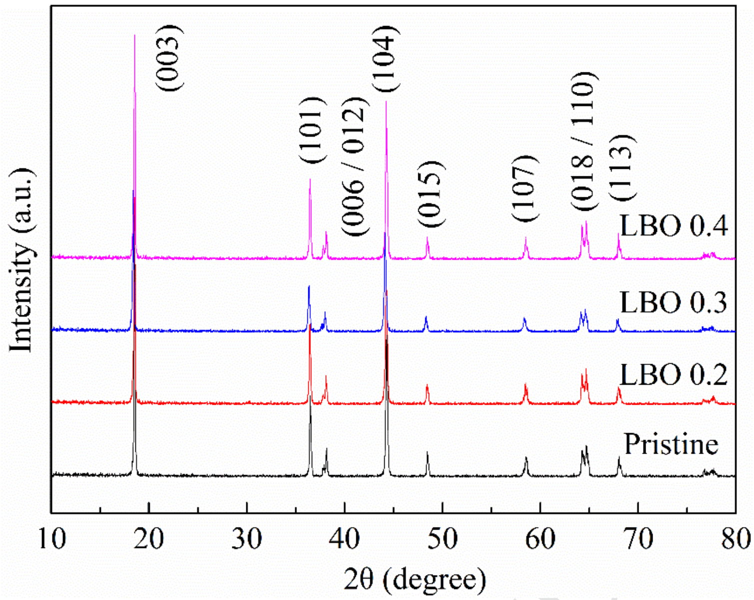

Figure 1. XRD patterns of pristine NCM811 powder and NCM811 powder coated with LBO glass: 0.2 wt% (LBO 0.2), 0.3 wt% (LBO 0.3), and 0.4 wt.% (LBO 0.4).

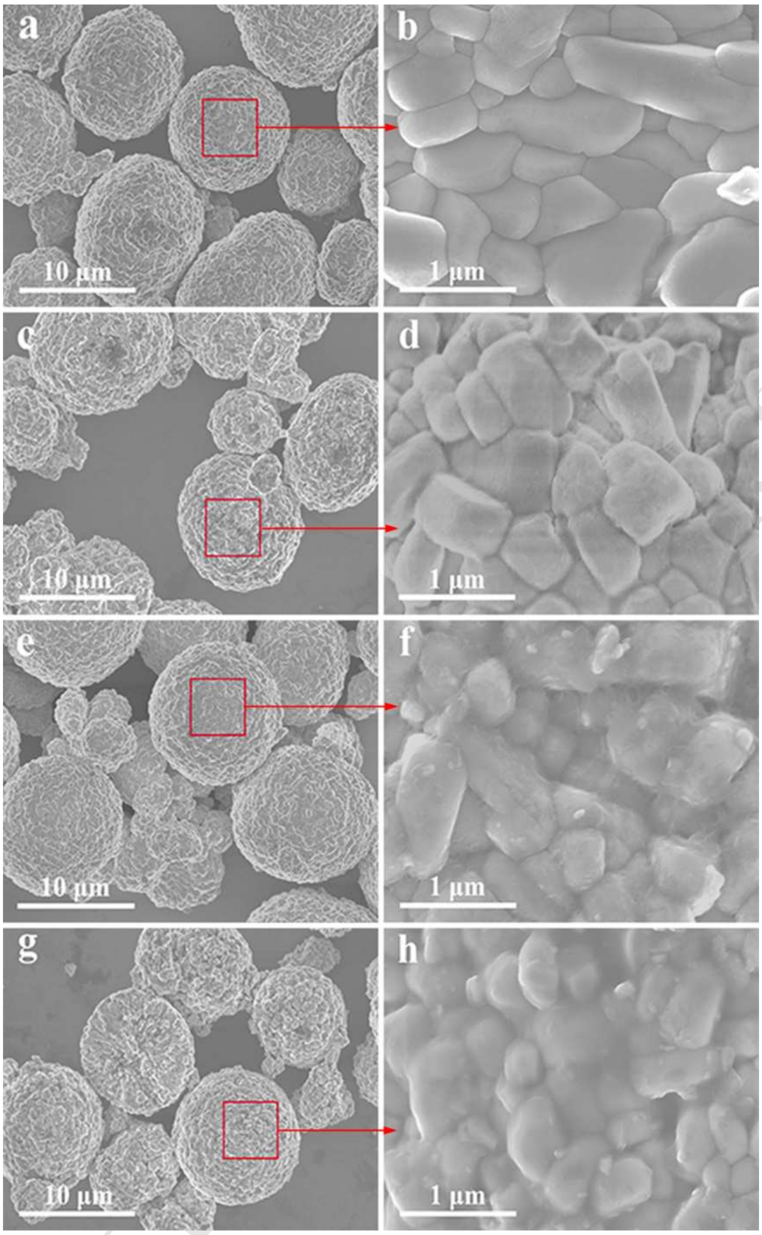

Figure 2. SEM images of pristine powder (a and b), LBO 0.2 (c and d), LBO 0.3 (e and f), and LBO 0.4 (g and h).

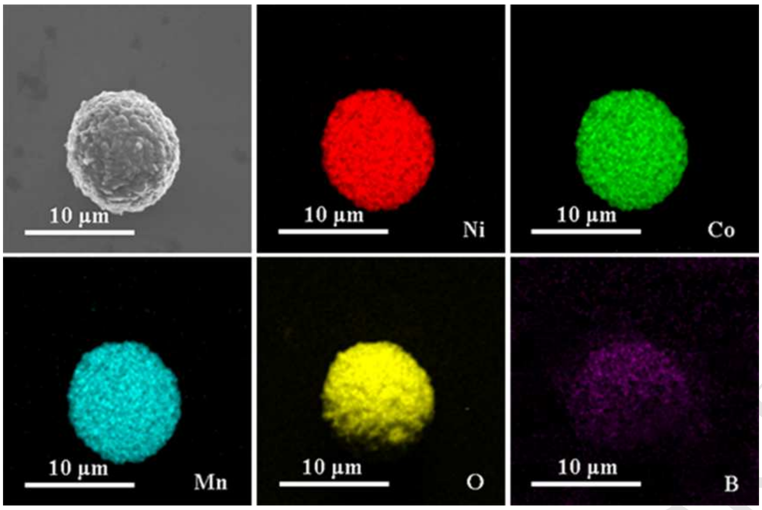

Figure 3. EDS elemental mapping of Ni, Co, Mn, O, and B for LBO 0.3 sample.

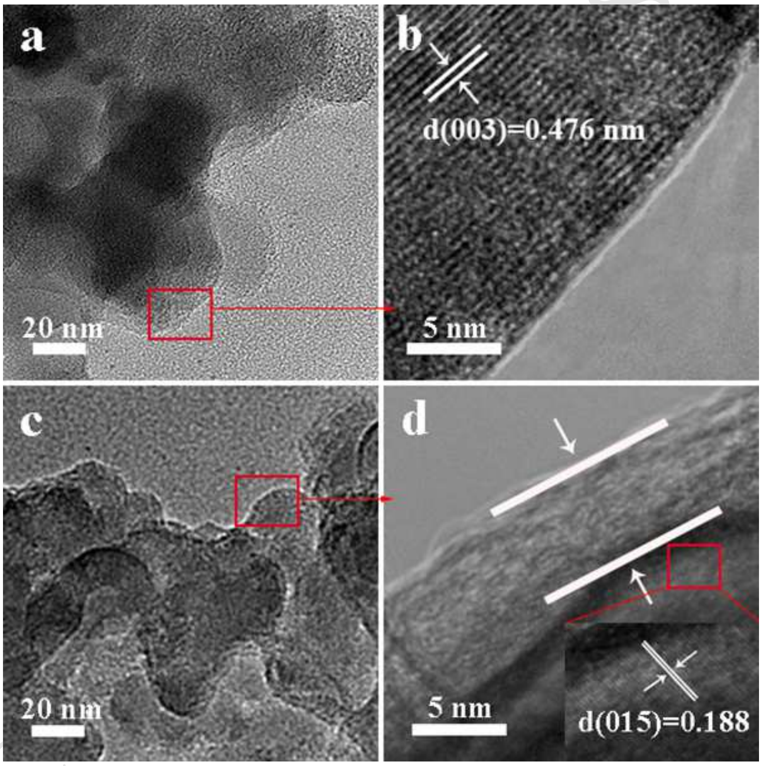

Figure 4. FE-TEM images of pristine NCM811 (a and b) and 0.3 wt% LBO-coated samples (c and d).

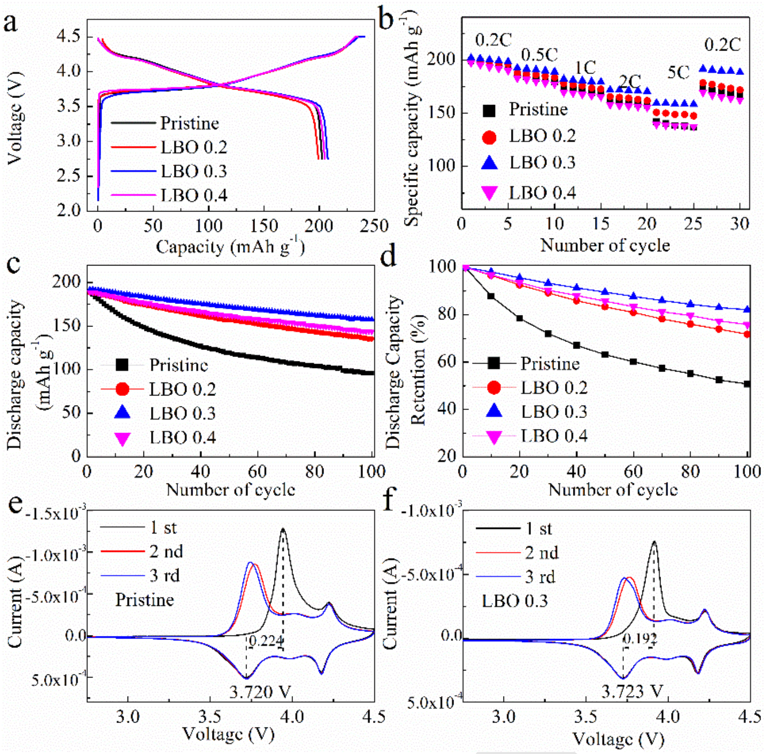

Figure 5. Electrochemical properties of pristine NCM811 and LBO 0.2, LBO 0.3, and LBO 0.4 samples in the voltage range of 2.75-4.5 V. (a) Initial charge/discharge curves at 0.1C (18 mAh g^{-1}). (b) Rate capability at various current densities. (c) Cycle performance at 1C (180 mAh g^{-1}). (d) Discharge capacity retention at 1C. CV curves of (e) pristine NCM811 and (f) LBO 0.3 at a scan rate of 0.1 mV s^{-1}.

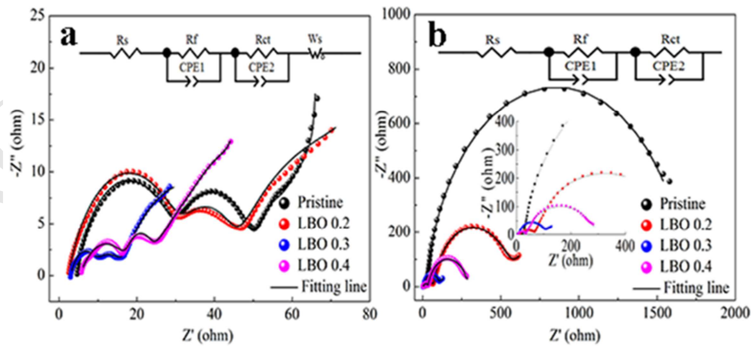

Figure 6. EIS results at (a) 1 cycle and (b) 100 cycles (after the 1st and 100th cycles in the voltage range of 2.75-4.5 V then charged to 3.8 V (1C)).

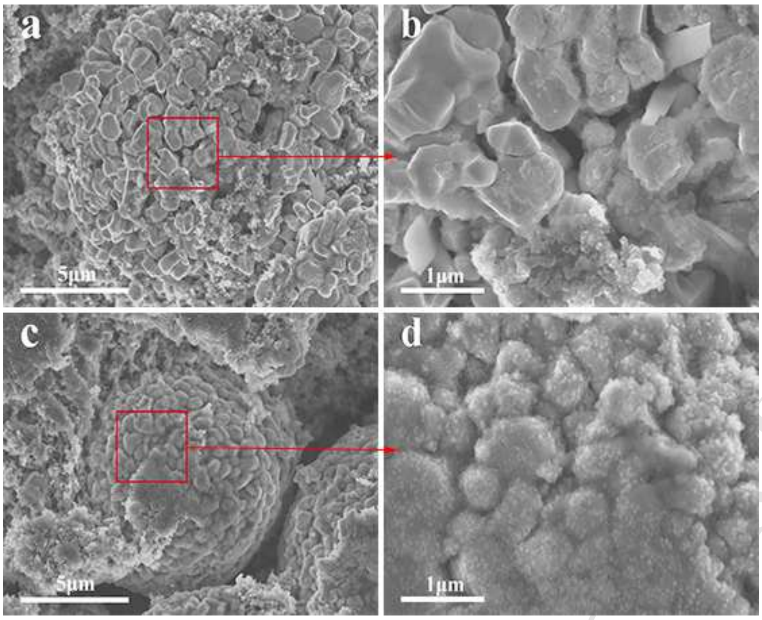

Figure 7. SEM images of electrodes after 100 cycles: (a-b) pristine NCM811 and (c-d) LBO 0.3.

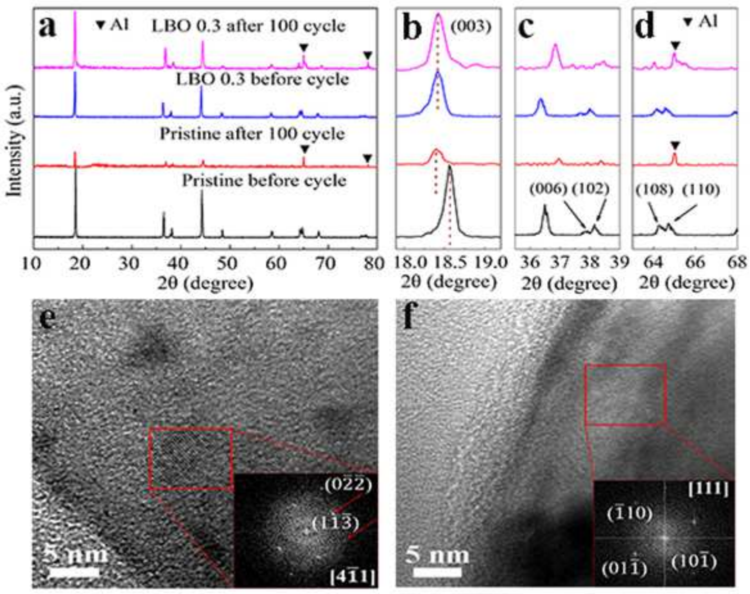

Figure 8. a-d) XRD patterns of pristine NCM 811 and LBO 0.3 before cycling and after 100 cycles. HR-TEM images and FFTs of (e) pristine NCM811 and (f) LBO 0.3 after 100 cycles.

$LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ is coated with a Li^+ conductive film of LBO-glass.

LBO coating facilitates the charge-transfer kinetics.

LBO-coated NCM811 exhibits superior cycling stability at high voltage of 4.5 V.

LBO coating alleviates surface structural degradation.